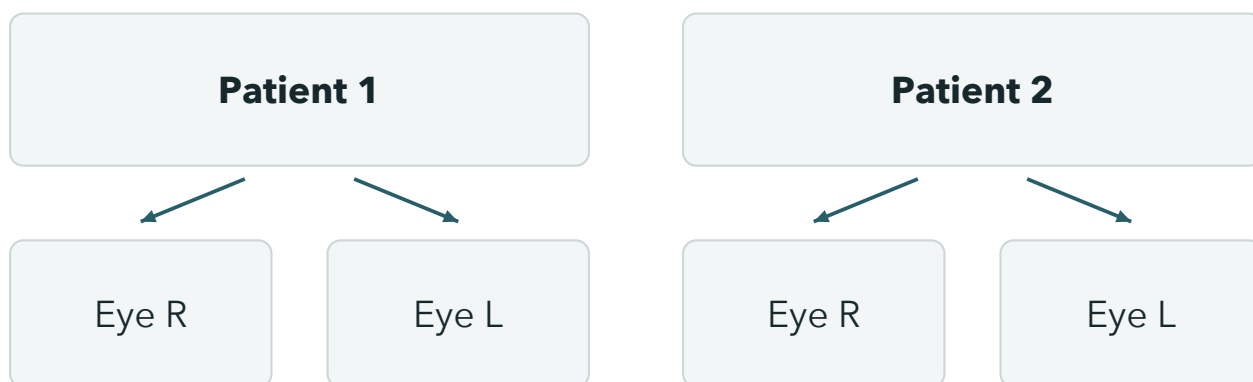


Trials, clustering and confidence intervals

Part 1 | Two eyes are not automatically two independent people

Positive fellow-eye correlation reduces the information gained by treating eyes as separate observations.



Fellow-eye outcomes share a patient-level cluster

Eye measurements are nested within a patient; the patient remains the shared cluster.

Uncertainty matters

If the same patient-level treatment is applied and both eyes are analysed as independent, uncertainty can be underestimated. Clustering need not reverse the treatment estimate to make its confidence interval invalid. The unit of randomisation, measurement and analysis must be described separately.

Choose an analysis that preserves the design

Paired-eye and patient-randomised designs have different dependence structures.

Design

Feature	Dependence	Analysis principle
Patients randomised; both eyes measured	Eyes clustered within patients	Account for clustering
One eye treated, fellow eye control	Within-patient pairing	Use paired structure
Repeated visits for each eye	Nested longitudinal data	Model relevant correlations

Avoid a shortcut

Discarding one eye is not the only valid solution; suitable correlated-data methods can use both. Conversely, doubling the nominal count without adjusting uncertainty does not create the information of twice as many independent patients.

Noninferiority uses a prespecified margin

The null boundary differs from the zero boundary used for superiority.

Higher outcome is better

Interval for new minus control	Conclusion
Entirely above -5, includes 0	Noninferiority supported; superiority not shown
Crosses -5	Noninferiority not established
Entirely above 0	Superiority may be supported under valid analysis

Original example

A difference of -1 with 95% CI -3 to +1 remains above a prespecified -5 margin but crosses zero. A small negative point estimate does not itself defeat noninferiority. The margin's clinical justification and trial conduct still matter.

Precision and importance are separate

A narrow interval does not make a clinically trivial effect important.

Question

Feature	Statistical aspect	Clinical aspect
How uncertain?	Confidence interval	Range of plausible effects
How large?	Effect estimate	Meaningful benefit or harm
For whom?	Study population	Applicability to patient

Evidence interpretation

Read the outcome scale before interpreting a numerical difference. A ten-letter acuity gain and a ten-percentage-point risk reduction have different meanings. The same p value can accompany very different effect sizes, and a non-significant estimate can still leave clinically important benefit or harm unresolved.

Check the design before the conclusion

A polished results table does not repair a mismatch between the biological design and the model.

Read an ophthalmic trial

- Identify randomisation unit and comparator
↓
- Locate paired or clustered observations
↓
- Check prespecified endpoint and margin
↓
- Assess uncertainty and applicability

An examination method

Separate a treatment-result question from an analysis-validity question. If the problem stipulates ignored correlation, address false precision first. If it supplies a prespecified noninferiority margin, compare the relevant interval bound with that margin instead of deciding from the p value for difference.